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Chapter 2

Functions

We know that God exists because Mathematics is consistent and we know that the devil exists because we cannot prove the consistency.

—Andre Weil

2.1 What is a Function?

A function is a *rule* or a *correspondence*, relating two sets in such a manner that each element in the first set corresponds to one and only one element in the second set. What do we mean when we say *y is a function of x?* Symbolically, we write $y = f(x)$, where

1. x is the *independent variable*. (input value of f)
2. y is the *dependent variable*. (output value of f at x)
3. f is a *function*. (rule that assigns x to y)

Definition 2.1.1. A function f from a set X to a set Y is a rule that assigns to each element x in X a unique element y in Y .

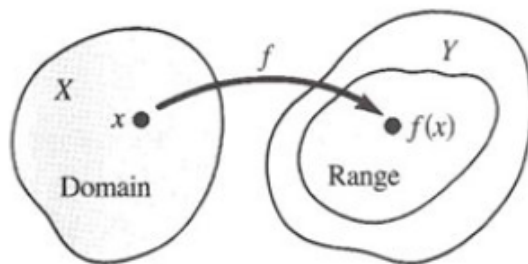
The set X is called the **domain** of f and the set of corresponding elements y in Y is called the **range** of f where sets X and Y consists of real numbers. We write $f : X \rightarrow Y$.

Examples: Stock market index depending on time, volume of sphere depending on radius, circle of a given radius has only one area.

Let f be a function. The number y in the range that corresponds to a selected number x in the domain is said to be the value of the function at x , or **image** of x , written $f(x)$, so $y = f(x)$.

Examples: $f(x) = 3x^4 + 2$, $g(t) = 4 - t^2$, $h(s) = 2s^2 + 7$.

The **domain** of a function f is the largest set of real numbers for which the rule makes sense.



Example: Let $f(x) = \frac{1}{x}$, we cannot compute $f(0)$, since $\frac{1}{0}$ is not defined. Then the domain of $f(x) = \frac{1}{x}$ is the set of all real numbers except 0.

Function	Domain $x \in X$	Range $y \in Y$
$y = x^2$	$(-\infty, \infty)$	$[0, \infty)$
$y = \frac{1}{x}$	$(-\infty, 0) \cup (0, \infty)$	$(-\infty, 0) \cup (0, \infty)$
$y = \sqrt{x}$	$[0, \infty)$	$[0, \infty)$
$y = \sqrt{1 - x^2}$	$[-1, 1]$	$[0, 1]$

Table 2.1: Examples of functions

You Try It: Let

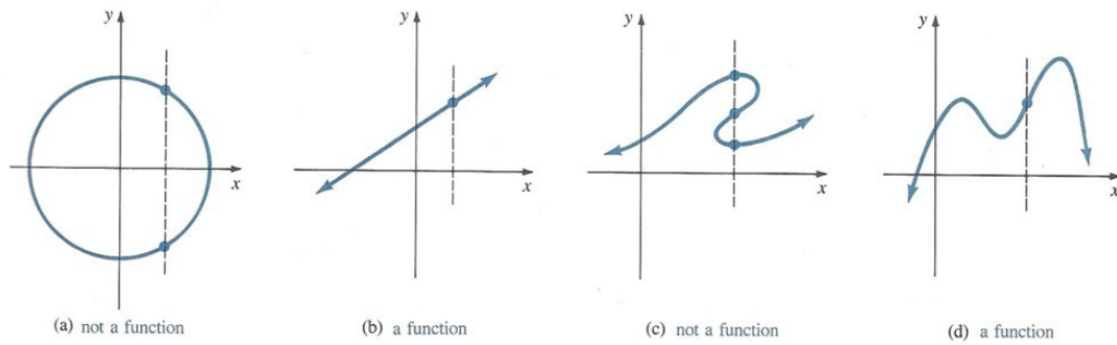
$$g(x) = \frac{x}{x^2 + 4x + 3}.$$

What is the domain and range of this function?

2.2 Graphs of Functions

It is useful to draw pictures which represent functions. These pictures, or *graphs*, are a device for helping us to think about functions. We graph functions in the $x - y$ plane. The elements of the domain of the function are thought of as points of the x -axis. The values of a function are measured on the y -axis. The graph of f associates to x a unique y value that the function f assigns to x . The **graph** of a function f is the set of points $\{(x, y) | y = f(x) \text{ in the domain of } f\}$ in the Cartesian plane.

As a consequence, a function is characterised geometrically by the fact that any vertical line intersecting its graph does so in *exactly one point*.



2.3 Bounded Functions

If there is a constant M such that $f(x) \leq M$ for all x in the interval (or other set of numbers), we say that f is *bounded above* in the interval (or the set) and call M an *upper bound* of the function. If a constant m exists such that $f(x) \geq m$ for all x in an interval, we say that $f(x)$ is *bounded below* in the interval and call m a *lower bound*. If $m \leq f(x) \leq M$ in an interval, we call $f(x)$ *bounded*.

Examples: $f(x) = x + 3$ is bounded in $-1 \leq x \leq 1$. An upper bound is 4 (or any number greater than 4). A lower bound is 2 (or any number less than 2).

2.4 Types of Functions

2.4.1 Elementary Functions

Polynomial Function

Have the form $f(x) = a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n$ where $a_0, a_1, \dots, a_n$ are constants and n is a positive integer called the degree of the polynomial provided $a_0 \neq 0$.

Examples: $x^5 + 10x^3 - 2x + 1$ is a polynomial of degree 5.

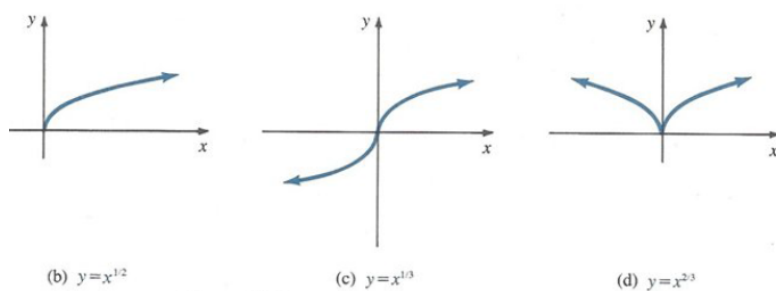
Rational Functions

A function $f(x) = \frac{P(x)}{Q(x)}$ where $P(x)$ and $Q(x)$ are polynomial functions.

Example: $f(x) = \frac{x^3 + x + 5}{x^2 - 3x - 4}$ is a rational function. Since $(x + 1)(x - 4)$ and $(x + 1)(x - 4) = 0$ for $x = -1$ and $x = 4$, the domain of f is the set of all real numbers except -1 and 4 .

Power Function

$f(x) = kx^n$, n a real number and k a constant.

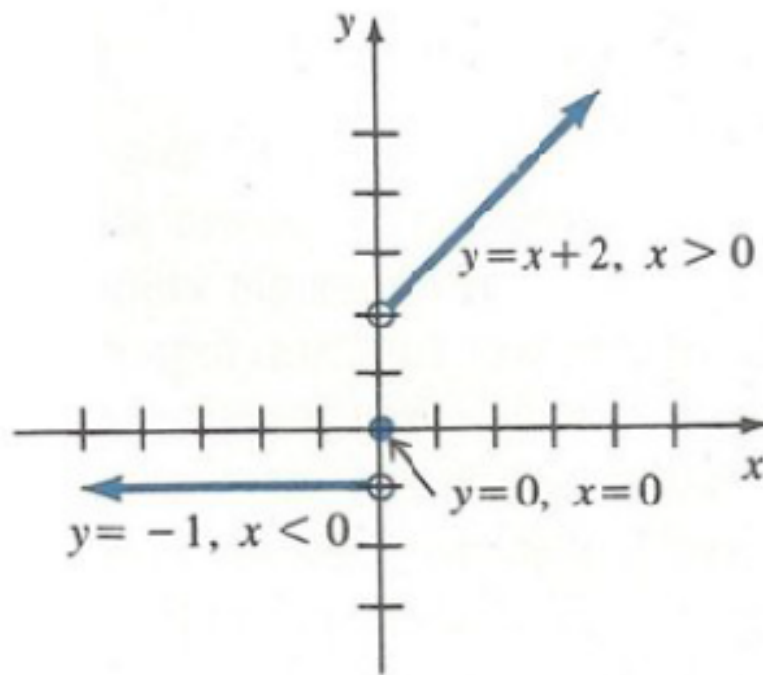


Examples: $y = \frac{1}{x}$, $y = x^{\frac{1}{2}}$, $y = x^{\frac{2}{3}}$.

Piecewise Defined Functions

A function need not be defined by a single formula. A piecewise defined function is a function described by using *different formula on different parts of its domain*.

Examples: (a) $f(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ x + 2, & x > 0 \end{cases}$ (b) $f(x) = \begin{cases} -x, & x < 0 \\ x^2, & 0 \leq x \leq 1 \\ 1, & x > 1. \end{cases}$



Transcendental Functions

The following are sometimes called *elementary transcendental functions*.

1. Exponential function, $f(x) = a^x, a \neq 0, 1$.
2. Logarithmic function, $f(x) = \log_a x, a \neq 0, 1$.
3. Trigonometric functions (also called circular functions because of their geometric interpretation with respect to the unit circle), e.g., $\sin x, \cos x, \tan x = \frac{\sin x}{\cos x}, \csc x, \cot x, \sec x$.
4. Inverse trigonometric functions, e.g., $y = \sin^{-1} x, y = \cos^{-1} x$.
5. Hyperbolic Functions, e.g., $\sinh x, \cosh x, \tanh x, \coth x$.

Even and Odd Functions

Let $f(x)$ be a *real-valued* function of a real variable. Then f is *even* if $f(x) = f(-x)$. (Symmetric with respect to the y -axis)

Examples: $|x|, x^2, x^4, \cos x, \cosh x$.

Let $f(x)$ be a real-valued function of a real variable. Then f is *odd* if $-f(x) = f(-x)$ or $f(x) + f(-x) = 0$. (Symmetric with respect to the origin)

Examples: $x, x^3, \sin x, \sinh x$.

Example: Determine whether the following function is odd or even $f(x) = \frac{3x}{x^2 + 1}$.

Solution:

$$f(-x) = \frac{3(-x)}{(-x)^2 + 1} = -\frac{3x}{x^2 + 1} = -f(x).$$

The function is odd.

2.5 Combining Functions

A function f can be combined with another function g by means of arithmetic operations to form other functions, the **sum** $f + g$, **difference** $f - g$, **product** fg and **quotient** $\frac{f}{g}$ are defined as :
Let f and g denote functions, then

1. Sum : $(f + g)(x) = f(x) + g(x)$.
2. Difference : $(f - g)(x) = f(x) - g(x)$.
3. Product : $(fg)(x) = f(x)g(x)$.
4. Quotient : $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$.

Example: If $f(x) = 2x^2 - 5$ and $g(x) = 3x + 4$. Find $f + g, f - g, fg, \frac{f}{g}$.

Solution:

$$\begin{aligned}(f + g)(x) &= (2x^2 - 5) + (3x + 4) = 2x^2 + 3x - 1. \\(f - g)(x) &= (2x^2 - 5) - (3x + 4) = 2x^2 - 3x - 9. \\(fg)(x) &= (2x^2 - 5)(3x + 4) = 6x^3 + 8x^2 - 15x - 20 \\ \left(\frac{f}{g}\right)(x) &= \frac{2x^2 - 5}{3x + 4}.\end{aligned}$$

2.6 Composition of Functions

Let f and g denote functions. The **composition** of f and g , written $f \circ g$ is the function $(f \circ g)(x) = f(g(x))$ and the composition of g and f , written $g \circ f$, is the function $(g \circ f)(x) = g(f(x))$.

Example: If $f(x) = x^2$ and $g(x) = x^2 + 1$, find $f \circ g$ and $g \circ f$.

Solution:

$$(f \circ g)(x) = f(g(x)) = f(x^2 + 1) = (x^2 + 1)^2 = x^4 + 2x^2 + 1.$$

and

$$(g \circ f)(x) = g(f(x)) = g(x^2) = (x^2)^2 + 1 = x^4 + 1.$$

In general, $f \circ g \neq g \circ f$.

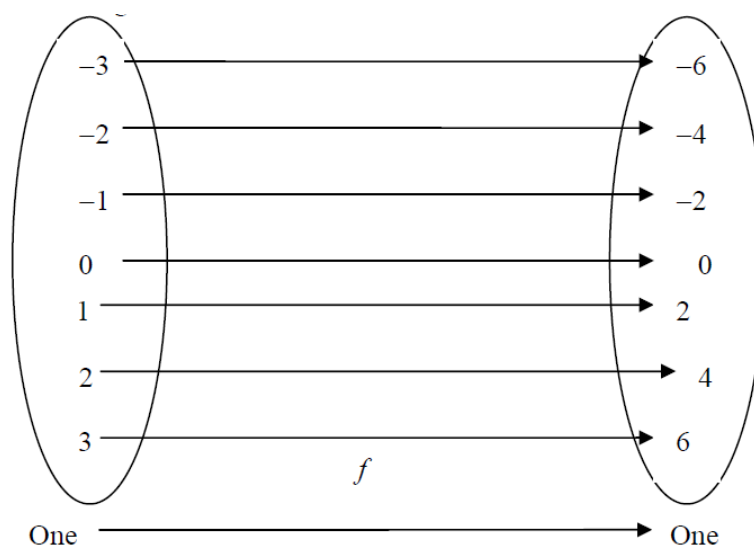
2.7 Bijection, Injection and Surjection

Classes of functions may be distinguished by the manner in which arguments and images are related or *mapped* to each other.

A function $f : X \rightarrow Y$ is **injective (one-to-one, 1-1)** if every element of the range corresponds to exactly one element in its domain X .

For all $x, y \in X$, $f(x) = f(y) \implies x = y$ or equivalently

For all $x, y \in X$, $x \neq y \implies f(x) \neq f(y)$. An injective function is an **injection**.



Example: Show that the functions $f(x) = 2x + 3$ and $g(x) = x^3 - 2$ are injective.

Solution: Need to show that $f(x) = f(y) \implies x = y$.

$$\begin{aligned}2x + 3 &= 2y + 3 \\2x &= 2y \implies x = y.\end{aligned}$$

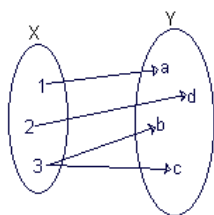
Hence $f(x) = 2x + 3$ is injective.

Need to show that $g(x) = g(y) \implies x = y$.

$$\begin{aligned}x^3 - 2 &= y^3 - 2 \\x^3 &= y^3 \quad \text{taking cube roots} \\x &= y.\end{aligned}$$

Hence $g(x) = x^3 - 2$ is injective.

A function $f : X \rightarrow Y$ is called **onto** if for all y in Y there is an x in X such that $f(x) = y$. All elements in Y are used. Such functions are referred to as **surjective**.



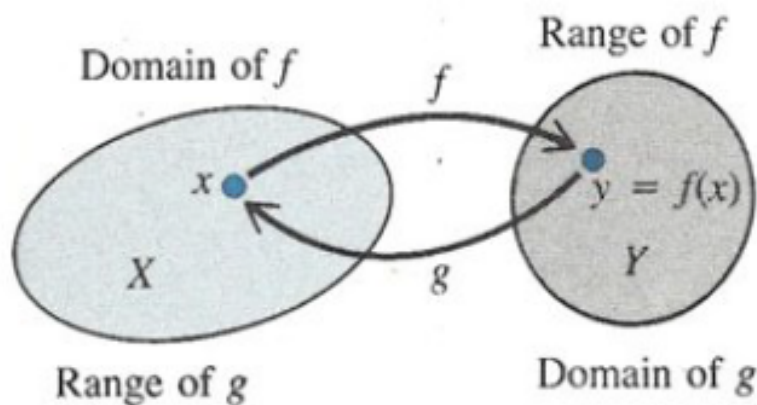
Example: Show that $f(x) = 3x - 5$ is onto.

Solution: For *onto* $f(x) = y$, i.e., $3x - 5 = y$. Solve for x , $\implies x = \frac{y + 5}{3}$.

So $f\left(\frac{y + 5}{3}\right) = 3\left(\frac{y + 5}{3}\right) - 5 = y$. Therefore f is onto.

Let X and Y be sets. A function $f : X \rightarrow Y$ that is one-to-one and onto is called a *bijection* or *bijjective function from X to Y* . If f is both one-to-one and onto, then we call f a 1 – 1 correspondence.

Inverse of a Function. Suppose f is a 1 – 1 function that has domain X and range Y . Since every element $y \in Y$ corresponds with precisely one element x of X , the function f must actually determine a *reverse* function g whose domain is Y and range is X , where f and g must satisfy $f(x) = y$ and $g(y) = x$. The function g is given the formal name *inverse* of f and usually written f^{-1} and read f inverse. Not all functions have inverses, those that do are called *invertible functions*.



Example: Find the inverse of the function $f(x) = (2x + 8)^3$.

Solution: We must solve the equation $y = (2x + 8)^3$ for x .

$$\begin{aligned} y &= (2x + 8)^3 \\ \sqrt[3]{y} &= 2x + 8 \\ \sqrt[3]{y} - 8 &= 2x \\ x &= \frac{\sqrt[3]{y} - 8}{2}. \end{aligned}$$

Hence the inverse function f^{-1} is given by $f^{-1}(x) = \frac{\sqrt[3]{x} - 8}{2}$.

A 1-1 function f can have only one inverse, i.e., f^{-1} is unique. A function $f : X \rightarrow Y$ is **invertible** if and only if f is one-to-one and maps X onto Y .

2.8 Operations on Functions

Equality of Functions

Equality of functions does not mean the same as equality of two numbers (numbers have a fixed value but values of functions vary). Each function is a relationship between x and y , the two relationships are the same if for every value of x we get the same value of y .

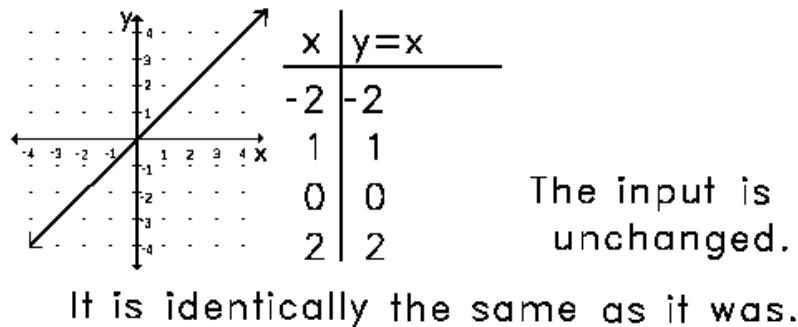
Example: The functions $(x - 1)(x + 2)$ and $x^2 + x - 2$ are equal.

Example: Equal functions for positive values of x ,

$$|x| = \sqrt{x^2}.$$

Identity Function

Generally, an identity function is one which does not change the domain values at all. Its the function $f(x) = x$. Denoted by I_X .



Monotonic Functions

A function is called *monotonic increasing* in an interval, if for any two points x_1 and x_2 in the interval such that if $x_1 < x_2$, $f(x_1) \leq f(x_2)$. If $f(x_1) < f(x_2)$ the function is called *strictly increasing*.

Similarly, if $f(x_1) \geq f(x_2)$ whenever $x_1 < x_2$, then $f(x)$ is *monotonic decreasing*, while if $f(x_1) > f(x_2)$ it is *strictly decreasing*.

2.9 Tutorial Questions

1. Find the domain of each of the following functions.

(i) $-\sqrt{9-x^2}$ (ii) $\frac{1}{\sqrt{1-x}}$ (iii) $\frac{1}{2x+3}$ (iv) $\sqrt{(3-x)(2x+4)}$ (v) $\frac{x-2}{x^2-4}$
 (vi) $\ln(x^3-3x^2-4x+12)$ (vii) $\frac{x^3-8}{x^2-4}$ (viii) $\sqrt{\sin 5x}$ (ix) $\ln(2x^2-5x-12)$
 (x) $\frac{1}{\sqrt{9-x^2}}$ (xi) $\sqrt{x^2-4}$ (xii) x^2+4 (xiii) $\sqrt{\frac{x}{2-x}}$.

2. If $f(x) = \frac{x-1}{x+1}$, find (i) $f(0)$ (ii) $f(1)$ (iii) $f(-2)$. Show that $f\left(\frac{1}{x}\right) = -f(x)$ and $f\left(-\frac{1}{x}\right) = -\frac{1}{f(x)}$.

3. If $f(x) = \frac{1}{x}$, show that $f(a) - f(b) = f\left(\frac{ab}{b-a}\right)$.

4. If $f(x) = x^2 - x$, show that $f(x+1) = f(-x)$.

5. If $y = f(x) = \frac{5x+3}{4x-5}$, show that $x = f(y)$.
6. Determine whether or not each of the following correspondences is a function.
 - (i) $x^2 + y = 1$ (ii) $x^2 y^2 = 5$ (iii) $x^2 y = 4$ (iv) $\{(1, 5), (2, 5), (5, 1)\}$.
7. For each of the following pairs of functions, calculate $f \circ g$ and $g \circ f$.
 - (i) $f(x) = \sin(x + 3x^2)$ and $g(x) = \cos(x^2 - x)$.
 - (ii) $f(x) = e^{x^2}$ and $g(x) = e^{-x^2}$.
 - (iii) $f(x) = x(x+1)(x+2)$ and $g(x) = (2x-3)(x+4)$.
8. Determine whether each of the pairs of the following functions are equal and if they are not equal, give where they are equal.
 - (i) $f(x) = \sqrt{x^2}$, $g(x) = (\sqrt{x})^2$ (ii) $f(x) = x^2$, $g(x) = x|x|$.
9. Show that if $ad - bc \neq 0$, then the function $f(x) = \frac{ax+b}{cx+d}$ is one-to-one and find its inverse, stating the domains of both the function and its inverse. What happens if $ad - bc = 0$?
10. Show that these functions are a one-to-one correspondence. Find their inverses.
 - (i) $f(x) = 2x + 3$ (ii) $g(x) = x^3 - 2$ (iii) $h(x) = (x-2)^2$ (iv) $k(x) = \sqrt[3]{x} + 7$.
11. Show that the function $f(x) = \ln(x + \sqrt{x^2 + 1})$ is an odd function and find the inverse function of $f(x)$.